Assignment 01

Question 01

In [77]:
```
file=open('students-marks.txt')
print('Roll','\t  Letter Grade','\t      Grade Point')
for marks in file:
    marks=marks.split()
    mark=(int(marks[1])+int(marks[2]))/2+int(marks[3])+int(marks[4])
    if mark>100:
        print('Invalid mark!')
    elif mark>=80:
        print(int(marks[0]),'\t\tA+','\t\t4.00')
    elif mark>=75:
        print(int(marks[0]),'\t\tA','\t\t3.75')
    elif mark>=70:
        print(int(marks[0]),'\t\tA-','\t\t3.50')
    elif mark>=65:
        print(int(marks[0]),'\t\tB+','\t\t3.25')
    elif mark>=60:
        print(int(marks[0]),'\t\tB','\t\t3.00')
    elif mark>=55:
        print(int(marks[0]),'\t\tB-','\t\t2.75')
    elif mark>=50:
        print(int(marks[0]),'\t\tC+','\t\t2.50')
    elif mark>=45:
        print(int(marks[0]),'\t\tC','\t\t2.25')
    elif mark>=40:
        print(int(marks[0]),'\t\tD','\t\t2.00')
    else:
        print(int(marks[0]),'\t\tF','\t\t0.00')
```

```
Roll      Letter Grade      Grade Point
80              A+              4.00
81              A-              3.50
82              A+              4.00
```

Question 02

In [78]:
```
file=open('regex-sum-42.txt')
number=[]
for line in file:
    line=line.strip()
    parts=line.split()
    for part in parts:
        if part.isdigit():
            number.append(float(part))
print(sum(number))
print(number)
```

```
445822.0
[3036.0, 7209.0, 4497.0, 6702.0, 8454.0, 7449.0, 3665.0, 7936.0, 9772.0, 7114.0, 956.
0, 2564.0, 8003.0, 1704.0, 3816.0, 6662.0, 5858.0, 7777.0, 6482.0, 9634.0, 8805.0, 712
3.0, 9703.0, 4676.0, 6373.0, 2834.0, 7221.0, 2981.0, 5415.0, 6872.0, 4806.0, 5460.0, 8
533.0, 3538.0, 9663.0, 8001.0, 9795.0, 8752.0, 1117.0, 5349.0, 4509.0, 1004.0, 9258.0,
4183.0, 4034.0, 3342.0, 3482.0, 8567.0, 1052.0, 8135.0, 5561.0, 517.0, 1218.0, 8877.0,
8062.0, 1720.0, 279.0, 2054.0, 801.0, 918.0, 8687.0, 7073.0, 1865.0, 7084.0, 2923.0, 6
3.0, 8824.0, 1079.0, 5801.0, 5047.0, 5.0, 2572.0, 5616.0, 171.0, 3062.0, 9552.0, 7655.
0, 829.0, 6096.0, 2312.0, 6015.0, 7100.0, 9548.0, 2727.0, 1792.0, 8402.0, 42.0]
```

Question 03

In [79]:
```python
word=[]
teachers=open('amth.txt')
for teacher in teachers:
    teacher.strip()
    words=teacher.split()
    for word_list in words:
        if word_list not in word:
            word.append(word_list)
print(word)
word.sort()
print(word)
```

```
['Applied', 'Mathematics', 'Honorable', 'Teachers:', 'Dr.', 'Mohammad', 'Ferdows', 'M
d.', 'Abdus', 'Samad', 'Shafiqul', 'Islam', 'Litan', 'Kumar', 'Saha', 'A', 'B', 'M',
'Shahadat', 'Hossain', 'Zavid', 'Iqbal', 'Bangalee', 'Tania', 'Sharmin', 'Khaleque',
'Rakib', 'Kajal', 'Chandra', 'S.M.', 'Chapal', 'Muntasir', 'Alam', 'Shuchi', 'Chaki',
'Nirab']
['A', 'Abdus', 'Alam', 'Applied', 'B', 'Bangalee', 'Chaki', 'Chandra', 'Chapal', 'D
r.', 'Ferdows', 'Honorable', 'Hossain', 'Iqbal', 'Islam', 'Kajal', 'Khaleque', 'Kuma
r', 'Litan', 'M', 'Mathematics', 'Md.', 'Mohammad', 'Muntasir', 'Nirab', 'Rakib', 'S.
M.', 'Saha', 'Samad', 'Shafiqul', 'Shahadat', 'Sharmin', 'Shuchi', 'Tania', 'Teacher
s:', 'Zavid']
```

Question 04

In [80]:
```python
file=open('mbox.txt')
count=0
total=0.0
for line in file:
    if line.startswith('X-DSPAM-Confidence:'):
        value=float(line.split(':')[1].strip())
        total+=value
        count+=1
if count>0:
    average=total/count
else:
    average=0.0
print(count)
print(f'Average spam confidence: {average:.4f}')
```

```
1797
Average spam confidence: 0.8941
```

Question 05

In [81]:
```
file=open('mbox.txt')
count=0
lists=[]
for line in file:
    if line.startswith('From '):
        count+=1
        value=str(line.split()[1].strip())
        if value not in lists:
            lists.append(value)
            print(value)
print(count)
```

```
stephen.marquard@uct.ac.za
louis@media.berkeley.edu
zqian@umich.edu
rjlowe@iupui.edu
cwen@iupui.edu
gsilver@umich.edu
wagnermr@iupui.edu
antranig@caret.cam.ac.uk
gopal.ramasammycook@gmail.com
david.horwitz@uct.ac.za
ray@media.berkeley.edu
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stuart.freeman@et.gatech.edu
tnguyen@iupui.edu
chmaurer@iupui.edu
aaronz@vt.edu
ian@caret.cam.ac.uk
csev@umich.edu
jimeng@umich.edu
josrodri@iupui.edu
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dlhaines@umich.edu
hu2@iupui.edu
sgithens@caret.cam.ac.uk
arwhyte@umich.edu
gbhatnag@umich.edu
gjthomas@iupui.edu
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bahollad@indiana.edu
jzaremba@unicon.net
mbreuker@loi.nl
colin.clark@utoronto.ca
1797
```

Question 06

In [82]:
```
file=open('mbox.txt')
count=0
lists={}
for line in file:
    if line.startswith('From '):
```

```
        count+=1
        value=str(line.split()[1].strip())
        if value in lists:
            lists[value]+=1
        else:
            lists[value]=1
print(count)
print(lists)
```

```
1797
{'stephen.marquard@uct.ac.za': 29, 'louis@media.berkeley.edu': 24, 'zqian@umich.edu':
195, 'rjlowe@iupui.edu': 90, 'cwen@iupui.edu': 158, 'gsilver@umich.edu': 28, 'wagnermr
@iupui.edu': 44, 'antranig@caret.cam.ac.uk': 18, 'gopal.ramasammycook@gmail.com': 25,
'david.horwitz@uct.ac.za': 67, 'ray@media.berkeley.edu': 32, 'mmmay@indiana.edu': 161,
'stuart.freeman@et.gatech.edu': 17, 'tnguyen@iupui.edu': 6, 'chmaurer@iupui.edu': 111,
'aaronz@vt.edu': 110, 'ian@caret.cam.ac.uk': 96, 'csev@umich.edu': 19, 'jimeng@umich.e
du': 93, 'josrodri@iupui.edu': 28, 'knoop@umich.edu': 5, 'bkirschn@umich.edu': 27, 'dl
haines@umich.edu': 84, 'hu2@iupui.edu': 7, 'sgithens@caret.cam.ac.uk': 43, 'arwhyte@um
ich.edu': 27, 'gbhatnag@umich.edu': 3, 'gjthomas@iupui.edu': 44, 'a.fish@lancaster.ac.
uk': 14, 'ajpoland@iupui.edu': 48, 'lance@indiana.edu': 8, 'ssmail@indiana.edu': 5, 'j
lrenfro@ucdavis.edu': 1, 'nuno@ufp.pt': 28, 'zach.thomas@txstate.edu': 17, 'ktsao@stan
ford.edu': 12, 'ostermmg@whitman.edu': 17, 'john.ellis@rsmart.com': 8, 'jleasia@umich.
edu': 2, 'ggolden@umich.edu': 8, 'thoppaymallika@fhda.edu': 1, 'kimsooil@bu.edu': 14,
'bahollad@indiana.edu': 4, 'jzaremba@unicon.net': 9, 'mbreuker@loi.nl': 9, 'colin.clar
k@utoronto.ca': 1}
```

In [ ]: